

Lock Box Capstone Project Report

1. Introduction (Who, What, When)

This project was completed by Dalton Hanna and Jorge Sosa as part of a mechatronics capstone course. The objective was to design and build a timed lock box system that restricts access to stored items for a user-defined period of time. The project was developed over the course of the semester and integrates mechanical design, electronics, and embedded programming.

2. Summary (Abstract)

This project presents the design and implementation of a low-cost, portable lock box system intended to improve user focus by limiting access to distractions such as smartphones. The system allows users to select a lock duration using a potentiometer, displays the selected time on an LCD, and locks the box using a servo motor once activated.

The design prioritizes simplicity, reliability, and affordability. An Arduino Nano processes user input and controls the locking mechanism using a non-blocking timer. Testing demonstrated reliable performance, accurate timing, and stable user input across multiple cycles.

3. Problem Definition

Many individuals struggle with distractions, especially from smartphones. Existing solutions are often expensive, not customizable, or easy to bypass.

Constraints:

- Portable (battery-powered)
- Low cost
- Simple operation
- Reliable performance

Budget: Designed using low-cost components with an estimated total under \$50.

4. Solutions Considered

Considered solutions included app-based locks, mechanical timers, and complex electronic systems.

App-based solutions were rejected due to ease of bypass. Mechanical timers lacked flexibility. Complex systems increased cost and difficulty.

The chosen Arduino-based solution provided simplicity, flexibility, and reliability.

Problems encountered included noisy input readings, servo alignment issues, and maintaining accurate timing.

5. Final Design

The system includes an Arduino Nano, servo motor, LCD display, potentiometer, push button, battery pack, and 3D printed enclosure.

Performance:

- Timer accuracy within ~5%
- Reliable operation across tests
- Stable input after filtering

Estimated Cost:

- Arduino: \$10
 - Servo: \$8
 - LCD: \$10
 - Potentiometer: \$2
 - Button: \$1
 - Battery: \$10
 - Enclosure: \$10
- Total: ~\$50

6. Conclusion and Moving Forward

The project successfully solves the problem by providing a reliable timed lock system.

Future improvements include adding a keypad, Bluetooth control, improved durability, and enhanced enclosure design.